

[Since students who enrolled during Academic year 2025-2026 did not study subject oriented VAC during their first year, such students will study following VAC course during their Fourth semester only for the A.Y.2025-2026. This course may be implemented for the First year students enrolled during Academic year 2025-2026 as their VAC course in semester-1 as course code 107.]

Course Code: 307

Course Title: Indian Knowledge System in context to Computers - 01

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Course Title	Indian Knowledge System in context to Computers - 01
Credits	2
Course Category	Value Added Course (VAC)
Level of Course	100-199 (Fundamental Level)
Teaching per Week	1 Hours Theory + 2 Hours of Lab/interactive work.
Minimum weeks/Hours per Semester	15 Hours Theory + 30 Hours of Practical (Including class work, examination, preparation etc.)
Review / Revision	-
Implementation Year:	A.Y. 2025-2026
Cognitive Skills of the Course	The purpose of this course is to provide a comprehensive understanding of the Indian Knowledge System, particularly in the fields of Mathematics and Astronomy. It aims to explore the ancient texts and sutras, such as the Lilavati Samhita, Suryasiddhanta, and Shulba Sutras, highlighting their profound contributions to mathematical principles, geometric operations, and astronomical theories. The course also emphasizes the application of Vedic Mathematics and its mental calculation techniques. By studying these ancient texts and their innovative methods, students will gain a deeper appreciation for India's historical contributions to science and mathematics, fostering critical thinking and analytical skills in these domains. Remembering: 1. Recall key mathematical and astronomical concepts from ancient texts (e.g., Lilavati Samhita, Suryasiddhanta). 2. List important sutras from Vedic Mathematics and ancient astronomical theories. Understanding: 1. Explain the significance of ancient mathematical techniques and astronomical principles. 2. Describe the application of Vedic Mathematics sutras like Nikhilam and Ekadhikena Purvena. Application: 1. Solve arithmetic, algebraic, and geometric problems using ancient Indian methods. 2. Use astronomical principles from Suryasiddhanta to predict eclipses and planetary motion. Analysis: 1. Compare ancient methods with modern mathematical and astronomical techniques. 2. Analyze the influence of Indian astronomy on later scientific developments. Evaluation: 1. Critique the effectiveness of Vedic Mathematics in modern problem-solving. 2. Evaluate the accuracy of ancient astronomical calculations against modern findings. Creation: 1. Develop original problems using Vedic Mathematics sutras. 2. Design a modern application of ancient Indian astronomical principles (e.g., space exploration).

Course Objective	1. To explore the mathematical concepts and techniques from ancient Indian texts like Lilavati Samhita and Suryasiddhanta. 2. To understand the principles of Vedic Mathematics and its application in modern problem-solving. 3. To analyze the contributions of ancient Indian astronomers like Aryabhata and Varahamihira in shaping early astronomical theories. 4. To examine the geometric and algebraic operations described in Shulba Sutras and their relevance to modern mathematics. 5. To develop a deeper appreciation for the richness and historical significance of the Indian Knowledge System in scientific advancements.									
Pre-requisite	Basic knowledge of English and Computer programming language (C/Python).									
Course Outcomes	CO1: To familiarize students with the ancient Indian mathematical concepts and techniques found in texts like Lilavati Samhita and Suryasiddhanta. CO2: To enable students to understand and apply Vedic Mathematics sutras for efficient problem-solving and mental arithmetic. CO3: To examine the contributions of Aryabhata and Varahamihira in ancient Indian astronomy and their impact on modern astronomical theories. CO4: To analyze the geometric and algebraic operations from the Shulba Sutras and explore their relevance in today’s mathematical applications. CO5: To foster an appreciation for the historical and scientific significance of the Indian Knowledge System in shaping the development of mathematics and astronomy.									
Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO)		PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PSO7	PSO8	
	CO1									
	CO2									
	CO3									
	CO4									
	CO5									
Course Content	Unit-1 : Indian knowledge system of Mathematics : 1.1 Ancient Indian Arithmetic from Lilavati Samhita by Bhaskarachary-I: 1.1.1 Arithmetic rule : Sutra (Verse 1) 1.1.2 Multiplication of Large Numbers: Sutra (Verse 5) 1.1.3 Division: Sutra (Verse 8): Unit-2 : Ancient Algebra and its implementation 2.1 Ancient Algebra and Geometry operations from Lilavati Samhita: 2.1.1 Algebra : Sutra (Verse 13) 2.1.2 Geometric Relationships: Sutra (Verse 17) 2.1.3 Understanding Lilavati Samhita theorem later taught as Pythagorean theorem (Geometry): Sutra (Verse 23) [Implementation of all sutras in computer Lab. Using C / Python / Any other Prog. Language.] Unit-3 : Indian knowledge system on Astronomy : 3.1 Ancient Indian Astronomy from Suryasidhdhanta by Aryabhata: 3.1.1 Motion of the Earth: Sutra (Verse 3.9) 3.1.2 Length of the Year : Sutra (Verse 3.10) 3.1.3 Lunar and Solar Eclipses: Sutra (Verse 4.5) 3.1.4 The Motion of Planets : Sutra (Verse 1.13) 3.1.5 The Influence of the Sun on Planetary Motion: Sutra (Verse 2.12) 3.1.6 Zodiac and Signs: Sutra (Verse 1.5) 3.1.7 Solar System: Sutra (Verse 1.15) 3.1.8 Speed of Planets: Sutra (Verse 6.5) 3.1.9 Planetary Distances from earth to moon: Sutra (Verse 7.8) 3.1.10 Latitude and Longitude of Planets : Sutra (Verse 8.12) Unit-4: Ancient Indian Astronomy and its Application: 4.1 Ancient Indian Astronomy by Varahmihir :									

	4.1.1 On Lunar Phases : Sutra (Verse 2.10) 4.1.2 On the Movements of the Stars : Sutra (Verse 2.18) 4.1.3 Ecliptic Latitude and Longitude 4.1.4 Sidereal and Tropical Years 4.1.5 Planetary Conjunctions and Aspects [Students will prepare a presentation on assigned topics and prepare a detailed report on given topic. Students will present the topic and submit the report as part of their final evaluation.]
Reference Books	1. "Vedic Mathematics", Bharati Krishna Tirthaji, Motilal Banarsidass Publishers, ISBN-13: 978-8170611552 2. "Lilavati", Bhaskaracharya (Bhaskara I), Oriental Publishers, ISBN-13: 978-8171101539 3. "Suryasiddhanta", Aryabhata, Motilal Banarsidass Publishers, ISBN-13: 978-8120818503 4. "The Shulba Sutras", J. L. Shastri, Motilal Banarsidass Publishers, ISBN-13: 978-8120817018 5. "A History of Ancient Indian Mathematics", S. N. Sen, Motilal Banarsidass Publishers, ISBN-13: 978-8120804247 6. "Mathematics in Ancient India", S. C. R. Anjaneyulu, Asia Publishing House, ISBN-13: 978-8120603404 7. "The Concept of Zero", Shukla S. K., M.D. Publications, ISBN-13: 978-8175332634 8. "Aryabhatiya of Aryabhata", Aryabhata, Varanasi: K. P. Jayaswal Research Institute, ISBN-13: 978-8185760255 9. "Indian Mathematics: History and Development", K. V. P. Subramanian, Springer, ISBN-13: 978-3319225829 10. "Indian Astronomy: A Study", M. A. Sastry, Cambridge University Press, ISBN-13: 978-0521270339
Teaching Methodology	Class Work, Discussion, Lab work, Self-Study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. : - Attendance, Class and home Assignment, - Lab work based on Unit-1,3 and 4; - Report writing and report presentation based on Unit-2. 50% External assessment. : (i) Practical exam to implement given problem(s) based on Unit-1,Unit-3 and Unit-4. (ii) Presentation on report prepared during the semester based on Unit-2.

Internship: Student willing to exit the program at the end of the two semesters and to avail the Certificate in Computer Application or exit the program at the end of the first four semesters and to avail the Diploma in Computer Application, it is essential to acquire four credits from internship. A key aspect of the internship is induction into actual work situations. Internships involve working with local industry, government or private organizations, business organizations, artists, crafts persons, and similar entities to provide opportunities for students to actively engage in on-site experiential learning. In option to these internships, the student can avail such four credits by availing two 2-credit university approved courses during any of these semesters. The student is required to enroll and avail these 4-credits and produce the evidence in process to opt the multi-level exit option after successfully completion of first year (two semester) or second year(four semesters).